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# Scaling Laws and Architectural Frontiers in Metagenomic Foundation Models

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## Abstract

Foundation models for genomics have the potential to revolutionize therapeutic design, yet the optimal architectural choices for modeling the vast and diverse distribution of metagenomic data remain under-explored. In this work, we present the machine learning methodology behind EDEN, a family of metagenomic foundation models scaled up to 28 billion parameters and trained on 9.7 trillion nucleotide tokens. We provide a systematic empirical study of architectural trade-offs between autoregressive Transformers (Llama-style), State-Space Models (Mamba), and Long-convolutional architectures (Hyena) for nucleotide-level modeling. Contrary to recent trends favoring linear-time sequence models for long-range biological data, we demonstrate that the Llama architecture exhibits superior scaling efficiency and semantic retrieval capabilities as the model capacity grows. We derive a set of quality-aware scaling laws for metagenomics, showing how model performance follows predictable power-law behavior across three orders of magnitude in parameters and data. Through extensive benchmarking, spanning unsupervised zero-shot fitness prediction, semantic completion, and gene recovery, we establish a blueprint for scaling biological foundation models and provide empirical evidence demonstrating why Transformer-based architectures define the current frontier.

## 1. Introduction

The recent rise of biological language models has reframed biological sequences as a substrate for general-purpose representation learning and generative modeling, enabling transfer and controllable generation across tasks, scales, and organisms. These models, pretrained on raw sequences with self-supervised objectives, produce reusable representations and priors that can be adapted to data-scarce downstream settings. In genomics, this paradigm has already proved effective for learning transferable representations that support a range of downstream tasks, including regulatory element prediction (e.g., promoters, splice sites, and transcription factor binding sites) with pretrained DNA models such as DNABERT and its multi-species successor DNABERT-2 (Ji et al., 2021; Zhou et al., 2024). Complementarily, large-scale genome-wide foundation models such as the Nucleotide Transformer further demonstrate transfer across diverse genomics benchmarks, including applications such as functional variant prioritization (Dalla-Torre et al., 2025).

Metagenomics (Handelsman, 2004) studies the collective genetic material of microbial communities, yielding sequences from multiple taxa with highly uneven abundances—a single gram of diverse environmental material can harbor thousands of species (Delmont et al., 2011), creating a heterogeneous, long-tailed distribution that challenges modeling. At scale, metagenomic assemblies yield tens of thousands of contigs from species absent in reference databases (Parks et al., 2017; Nayfach et al., 2021), demonstrating that metagenomic sequence space far exceeds curated collections. Foundation models trained on this diversity have begun to demonstrate practical value in variant effect prediction, generation across biological modalities (Nguyen et al., 2024), and tracking viral evolution (Zvyagin et al., 2022), with applications in agriculture and human health (Liu et al., 2025; Munsamy et al., 2026). Metagenomic foundation models present a clear opportunity to act as general engines for (i) representation learning across microbial sequence space, (ii) scalable annotation of “microbial dark matter,” and (iii) sequence generation to support downstream discovery pipelines.

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However, the architectural and design choices that best support scaling in metagenomics remain under-explored. Recent work has proposed linear-time alternatives to attention, demonstrating million-token context at nucleotide resolution (Nguyen et al., 2023). Yet it remains unclear which model family provides the best trade-off between scaling efficiency, the ability to capture long-range dependencies, and practical training throughput at trillion-token regimes.

We address this gap with a controlled empirical study comparing three sequence-modeling families - autoregressive Transformers, state-space models, and long-convolutional architectures - trained across multiple scales on a large metagenomic corpus and evaluated on nucleotide-level likelihood and long-context behavior. Building on these results, EDEN was developed, a family of metagenomic foundation models scaled up to 28 billion parameters and trained on up to 9.7 trillion nucleotide tokens from BaseData. The main contributions of this paper are:

- **Systematic architectural comparison:** We evaluate autoregressive Transformers (Llama), state-space models (Mamba), and long-convolutional architectures (Hyena) across three orders of magnitude in size (100M, 1B, 7B parameters). We find that while all architectures improve with scale, Transformers exhibit superior scaling efficiency on information-dense biological tasks.
- **Scaling laws and data quality:** We derive scaling laws for metagenomic language models, demonstrating predictable power-law behavior. Crucially, we quantify the impact of data quality: models trained on our curated, long-read-optimized dataset exhibit a significantly steeper scaling exponent than those trained on public short-read data, highlighting the critical role of assembly quality and diversity.
- **Semantic vs. structural coherence:** We uncover a fundamental trade-off in long-context generation. While linear-time models extrapolate structural properties (e.g., coding density) more gracefully beyond the training window, they generate sequences that diverge semantically from expected biological continuations. In contrast, Transformers maintain high semantic fidelity - correctly predicting downstream operon genes - demonstrating that global attention is essential for capturing the biological logic of gene regulation.

We emphasize that the primary goal of this work is not to provide an exhaustive benchmark of the final EDEN-28B model, but rather to detail the systematic scaling analysis and architectural insights that guided its development.

The remainder of this paper is organized as follows. Section 2 reviews related work on genomic foundation models. Section 3 details our data curation process and the experimental setup, including the architectural paradigms eval-

uated. Section 4 presents the comparative benchmarking results and derives scaling laws for metagenomics. Section 5 examines context length extrapolation and semantic coherence. Section 6 describes the scaling of the EDEN model to 28B parameters. Finally, we conclude with limitations and future directions.

## 2. Related Work

**Encoder-based DNA models.** The development of genomics foundation models has broadly mirrored trends in NLP, moving from encoder-style pretraining toward large generative models. Early efforts such as DNABERT (Ji et al., 2021) adapted masked-language modeling to  $k$ -mer tokenized DNA and demonstrated transfer to discriminative genomics tasks including promoter, splice-site, and TF-binding prediction. DNABERT-2 (Zhou et al., 2024) extends this line of work to multi-species settings while replacing fixed  $k$ -mers with a learned BPE tokenizer and introducing a standardized benchmark for genome understanding. Along similar lines, the Nucleotide Transformer (Dalla-Torre et al., 2025) scales masked pretraining to models up to 2.5 billion parameters trained on broad genomic corpora, yielding transferable representations that support a wide range of downstream analyses, including variant-centric evaluations.

**Regulatory genomics models.** Long-context sequence-to-function predictors have advanced rapidly in regulatory genomics. Enformer (Avsec et al., 2021) uses self-attention over 200 kb input windows to predict functional genomics tracks, and AlphaGenome (Avsec et al., 2025) extends it to multimodal prediction from input sequences up to 1 Mb.

**Generative genomic architectures.** A line of work explores generative architectures that learn nucleotide-level sequence priors for *de novo* generation and design. Evo-1 and Evo-2 (Nguyen et al., 2024; Brixi et al., 2025) demonstrate long-context genomic modeling and generation using the Striped Hyena family of hybrid convolutional architectures, emphasizing the role of scalable long-range sequence operators. HyenaDNA (Nguyen et al., 2023) shows that long-convolutional models can reach megabase-scale context at single-nucleotide resolution with models up to 6.6M parameters. Beyond convolutions, state-space models have also been adapted to genomics: the MambaDNA block introduced in (Schiff et al., 2024) achieves strong performance on long-range variant-effect prediction benchmarks.

**Scaling laws for language models.** The discovery that neural language model performance follows predictable power-law scaling with compute, data, and parameters has transformed large-scale model development (Kaplan et al., 2020). Hoffmann et al. (2022) established compute-optimal training recipes, and Henighan et al. (2020) extended scaling analysis to diverse modalities. In biology, scaling laws have

been studied for protein language models (Lin et al., 2023) and recently for genomic models (Nguyen et al., 2024), though the latter was limited to 1B parameters.

**Metagenomic corpora and foundation models.** Recent work has explored foundation-model training directly on metagenomic data. While (Zvyagin et al., 2022) demonstrate the utility of language models for viral evolution, their design is heavily tailored to SARS-CoV-2, limiting generalization to broader biological classes. Closer to our approach, (Liu et al., 2024) adapt the Llama architecture for genomic data; however, their reliance on short-read sampling restricts their model to a narrow context window of 512bp. Recent work by (Zhou et al., 2025) has shown that scaling autoregressive architectures on metagenomic assemblies—from 100M to 4B parameters yields significant gains in data representation quality and generation. Finally, (Nguyen et al., 2024) established a baseline scaling law comparing Transformer++, Mamba, and Hyena architectures. Crucially, however, their benchmarks were restricted to the 1B parameter regime, leaving the comparative behavior of these architectures at larger scales unexplored.

### 3. Data and experimental setup

#### 3.1. Data sourcing and curation

Public genomic repositories exhibit significant taxonomic unevenness, heavily skewing toward clinically relevant organisms while leaving vast swathes of phylogenetic diversity underexplored (Hernandez et al., 2020). These data disparities propagate into downstream applications, creating biases that limit the generalization capabilities of language models by overfitting to well-sampled lineages (Ding & Steinhardt, 2024). To overcome these limitations, EDEN models are trained on BaseData explicitly curated to counterbalance this redundancy. Unlike standard repositories dominated by isolates, our corpus is enriched for environmental and host-associated metagenomes, allowing us to capture the “microbial dark matter” of complex communities. The full training corpus consists of approximately 9.7 trillion nucleotide tokens, representing a fundamental shift towards a more comprehensive representation of biological sequence space (see Appendix B for further details).

We apply stringent filtering criteria to ensure high data quality: contigs must exceed 2 kb in length, exhibit gene density  $>20\%$ , and have sequencing depth  $\geq 4\times$ . Low-complexity and eukaryotic-viral sequences were excluded (see Appendix B for full filtering details).

#### 3.2. Tokenization strategy

To model DNA at its native resolution, we employed a byte-level tokenizer. While k-mer tokenization reduces sequence length, it obscures the single-nucleotide variants that often

drive biological function. EDEN tokenizes DNA sequences at single-nucleotide resolution with a vocabulary size of 512, which accommodates the four canonical nucleotides (A, C, G, T), special tokens (BOS, EOS, SEP, PAD), and additional reserved slots to accommodate control tags for downstream finetuning. This approach allows the model to generate valid sequences character-by-character without the inductive bias or tokenization artifacts associated with fixed k-mers.

#### 3.3. Training pipeline

Data was processed using a sliding window approach. Genomic sequences of arbitrary length were partitioned into overlapping windows of 8,192 tokens with a stride of 7,992 tokens, providing a 200 bp overlap to preserve local context at window boundaries. Each window includes a Beginning-Of-Sequence (BOS) token, a Separator (SEP) token and ends with an End-Of-Sequence (EOS) token. For comparability, all models ranging from 100M to 7B parameters were trained on randomly sampled subsets matched for a total nucleotide budget of 350 billion nucleotides per dataset. The same random seed was used to ensure that all models were exposed to identical data samples.

#### 3.4. Model architectures and training

To inform the design of the EDEN-28B model and derive rigorous scaling laws, we evaluated three prevailing architectural paradigms across three orders of magnitude in size (100M, 1B, and 7B parameters), with an additional 300M Llama variant to provide a finer-grained intermediate point for the scaling-law fit:

- Llama3 (Transformer): An autoregressive baseline utilizing standard global attention ( $O(N^2)$  complexity).
- Mamba2 (SSM): A selective State-Space Model offering linear-time scaling ( $O(N)$ ) with sequence length.
- Hyena: A long-convolutional architecture optimized for long-context modeling.

The Mamba and Hyena architectures we train are hybrid architectures (Brix et al., 2025) consisting of interleaved attention and Mamba or Hyena layers, respectively. Specifically, we employ the Mamba-2-Hybrid and Striped Hyena configurations, where approximately one attention layer is interleaved for every seven recurrent/convolutional layers. This hybrid design reflects standard practice in the field, as pure linear-time models have been shown to underperform hybrids at scale (Waleffe et al., 2024). For brevity, we refer to these hybrid architectures as “Mamba” and “Hyena”, highlighting the key architectural differentiator. All models were trained on identical subsets of our BaseData dataset or OpenGenome2 (OG2), for up to 350 billion tokens. We note that at equivalent parameter counts, architectures differ in FLOPs due to the computational cost of attention

versus recurrent/convolutional operations (see Table 5); we discuss the implications of this in Section 4. We defer to Appendix C for further details on the training protocol.

**Bootstrap evaluation.** To assess the seed-to-seed stability of the test perplexities used in our scaling-law analysis, we re-evaluated each Llama checkpoint six times with independent evaluation seeds ( $\{42, 29, 123, 456, 789, 1024\}$ , identical across every model and dataset). Final perplexities are reported as the mean over the six runs; the seed-to-seed variation is small in every configuration ( $\sigma \leq 0.043$ ), and the qualitative ordering of the scaling exponents for Base-Data versus OpenGenome2 is preserved across all bootstrap replicates.

## 4. Scaling laws for metagenomics

Genomic modeling demands architectures that can efficiently resolve complex, non-local dependencies over long contexts. To identify the optimal architecture for this task, we evaluated the three candidate architectures described in Section 3 across three orders of magnitude in size (100M, 1B, and 7B parameters).

### 4.1. Benchmarking results: gene autocompletion

Gene autocompletion provides a stringent and biologically meaningful test of a genomic language model’s ability to learn the “grammar” of DNA sequences. Successful completion requires the model to capture both local sequence constraints, such as codon structure and conserved motifs, and longer-range dependencies that span functional domains and gene-scale context. Unlike short-range prediction tasks, autocompletion probes whether the model can generate coherent, functionally plausible sequence continuations, reflecting an internalised representation of gene architecture and evolutionary conservation.

To evaluate autocompletion, we prompted each model with the first 20% and 30% of a coding sequence and generated the remaining 80% and 70% autoregressively. We evaluated performance on three highly conserved housekeeping genes: *ftsZ* (cell division), *recA* (DNA repair), and *secY* (protein

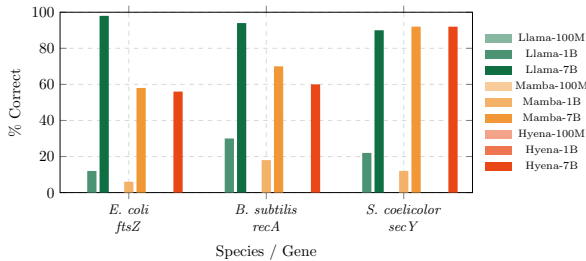


Figure 1. Percentage of generated sequences with correct domain architecture. Note that some of the models are not able to generate the correct domain at all.

translocation), across three model organisms (*E. coli*, *B. subtilis*, *S. coelicolor*). Generated sequences were evaluated using Pfam annotations (Paysan-Lafosse et al., 2025) and TM-scores (normalised to reference length) between ESM-Fold structures of proteins translated from the generations and wild-type proteins using USalign (Zhang et al., 2022).

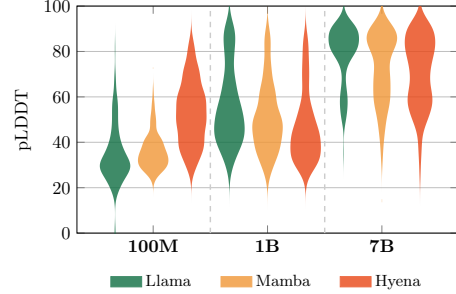


Figure 2. Distribution of pLDDT scores.

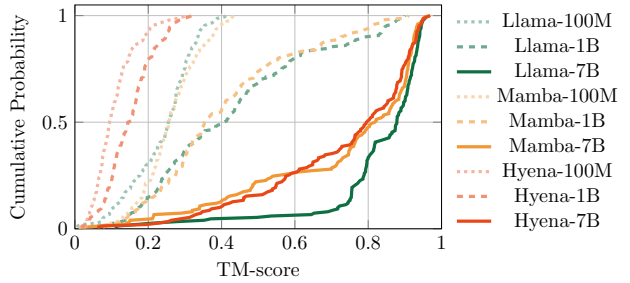


Figure 3. Empirical cumulative distribution of TM-scores.

Across organisms and genes, smaller models showed limited generative capability, often failing to recover the correct domain organization, whereas 7B-parameter models substantially improved domain-architecture recovery as seen in Figure 1. This scaling effect is also reflected in structural predictions: Llama-based models produced the most reliable structures, with Llama-7B showing higher pLDDT values and closer agreement with expected folds in Figure 2, consistent with a marked increase in predicted structural confidence relative to smaller models which most often yielded low-confidence structures. Furthermore, the TM-score distribution in Figure 3 is right-shifted for Llama, indicating closer agreement with reference folds and fewer failures (low TM-scores) compared to Mamba and Hyena equivalents. This trend holds across all three genes. In Appendix E, we provide an additional structural evaluation, including a breakdown of TM-scores by gene and the correlation between sequence identity and structural accuracy. Crucially, we observe that Llama maintains high structural fidelity even in regimes of low sequence identity. This indicates that the model captures biological semantics and structural constraints over mere sequence memorization.

Together, these results suggest that model capacity strongly influences biologically coherent gene completion and moti-

vates our subsequent architecture selection for large-scale training.

#### 4.2. Information compression vs. model capacity

Global metrics such as average perplexity are standard proxies for capacity in scaling law experiments, yet they often obscure local performance nuances. To better assess the semantic capacity of our models, we evaluate how each architecture copes with increasing functional density, using Kyoto Encyclopedia of Genes and Genomes (KEGG) annotations as a ground truth for biological signal (Kanehisa et al., 2016). We stratify the test set along two distinct axes:

- **Functional Density (Complexity):** We first measure how sequence-level perplexity evolves with the *count* of KEGG annotations present in a sequence. This tests how model design and scale influence the resolution of semantically rich, information-dense regions.
- **Annotation Rarity (Generalization):** We then measure how perplexity correlates with the *frequency* of specific KEGG annotations within the training corpus. This tests the model’s ability to recall and generalize to rare or under-represented genetic elements.

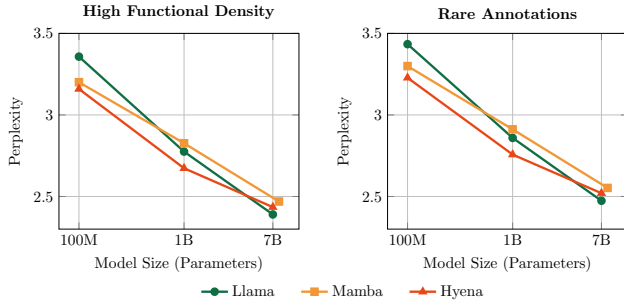


Figure 4. Information compression vs. model capacity. We evaluate architectural performance on (left) sequences with high functional density ( $\geq 10$  KEGG annotations) and (right) sequences containing rare annotations ( $< 10^3$  occurrences in training).

When analyzing scaling behavior through the prism of functional density in Figure 4, we observe that Llama-based models exhibit superior scaling dynamics compared to their linear-time counterparts (Hyena and Mamba). While all architectures improve with size, Llama benefits most significantly from increased capacity, ultimately achieving best-in-class performance at the 7B parameter scale.

This architectural advantage is further supported by our generalization analysis. When stratifying sequences by the rarity of their KEGG annotations, we observe a distinct advantage for Llama on both rare and more common genetic elements, indicating that global attention is critical for resolving biological signals.

#### 4.3. Power-law behavior and data quality

We observe that model performance follows a predictable power-law relationship between test loss ( $L$ ) and compute ( $C$ ), of the form  $L(C) \propto C^{-\alpha}$ . Following established methodology (Kaplan et al., 2020; Hoffmann et al., 2022), we fit scaling laws in log-log space using ordinary least squares regression on validation loss at convergence. To guard against single-seed noise, each Llama checkpoint used in the scaling-law fit was re-evaluated under six independent evaluation seeds (the same seed set across every model and dataset) and the bootstrapped mean test perplexity is used for the regression; full protocol details are reported in Section 3.

To isolate the impact of data quality on model performance, we conducted a controlled scaling analysis, benchmarking our BaseData corpus against the OpenGenome2 baseline (Brixi et al., 2025). Crucially, we observe that the scaling exponent  $\alpha$  is indeed data-dependent. Models trained on our in-house dataset follow a steeper power-law trajectory ( $\alpha \approx 0.059$ ) compared to those trained on OpenGenome2 ( $\alpha \approx 0.047$ ), see Figure 5. This difference indicates that

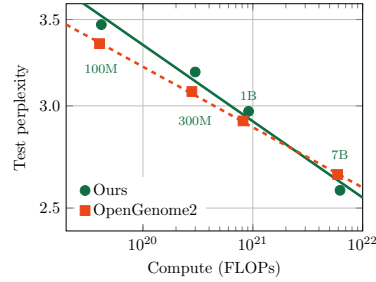


Figure 5. Scaling laws. Models trained on our BaseData dataset (green, solid) exhibit a steeper scaling exponent compared to models trained on public OpenGenome2 data. Data points correspond to 100M, 300M, 1B, and 7B parameter Llama models.

our dataset possesses a higher density of learnable signal, yielding superior performance gains for every unit of added model capacity.

We attribute this advantage to three key factors. First, our assembly pipeline is optimized for long-read sequencing, yielding a mean contig length of  $\approx 18$  kb, significantly exceeding the  $\approx 4$  kb average of OpenGenome2. This extended contiguity enables the model to resolve long-range structural dependencies, such as operon architecture and viral integration sites, which are frequently disrupted in short-read assemblies. Second, this reduced fragmentation directly minimizes “edge effects” during tokenization. By preserving the physical continuity of the sequence, we ensure that functional units - such as *cis*-regulatory elements and their target genes - are consistently presented within the same context window, facilitating the learning of gene regulation logic. Finally, our sampling strategy actively mitigates the bias toward well-studied reference genomes. By enriching for underrepresented taxa from diverse environments,

we generate a training distribution that is less dominated by clinical isolates and standard laboratory strains, providing a richer signal for generalization at larger scales.

**Statistical considerations.** We acknowledge that fitting power laws to three data points provides limited statistical power. However, we note that the fits exhibit high linearity in log-log space ( $R^2 > 0.985$  for both datasets). Note also that the training dynamics supporting these observations are detailed in Appendix D, where loss trajectories reveal that while architectures perform comparably at smaller scales, a clear performance separation emerges at 7B parameters, with Llama achieving distinctly lower validation loss. This scale-dependent crossover provides additional evidence for the architectural conclusions beyond the aggregate scaling exponents. Finally, note that we present these laws as empirical observations that guided our architectural decisions rather than as statistically definitive claims. Future work at additional scales would strengthen these estimates.

**Computational considerations.** As shown in Table 5, architectures differ in FLOPs at equivalent parameter counts due to the computational cost of attention versus recurrent/convolutional operations. At 7B, Mamba-Hybrid requires  $\sim 2.2\times$  the FLOPs of Llama, while Striped Hyena requires  $\sim 0.9\times$ . This reflects inherent architectural differences rather than implementation inefficiency. Our primary comparison matches *parameter count* and *token budget*, which controls for capacity and data exposure. We view the FLOPs difference as part of each architecture’s fundamental trade-off: linear-time models might offer better asymptotic scaling with sequence length but may require more operations per token to achieve equivalent expressivity.

#### 4.4. Zero-shot fitness prediction

Foundation models have demonstrated strong zero-shot fitness prediction when sequence likelihood is used as a proxy for functional activity (Bhatnagar et al., 2025; Nguyen et al., 2024; Dalla-Torre et al., 2025; Bixi et al., 2025). We evaluate this capability on the prokaryotic subset of RNAGym (Arora et al., 2025), which contains deep mutational scans covering millions of variants. As illustrated in Figure 6, Spearman correlation with measured fitness scales with model perplexity, with Llama excelling at the 7B scale.

## 5. To the context length and beyond

### 5.1. Long-context generation

We employ a context window of 8,192 tokens, sufficient to capture bacterial operons (genes average  $\sim 1000$  base pairs (bp) (Xu et al., 2006), operons  $\sim 3$  genes (Nuñez et al., 2013)) while enabling controlled cross-architecture com-

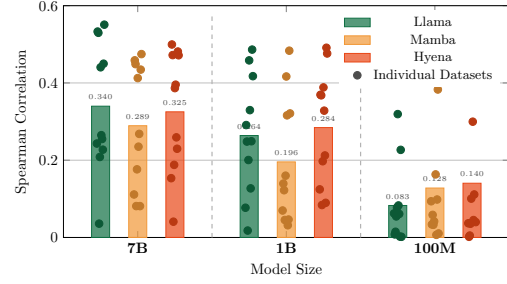


Figure 6. Zero-shot fitness prediction on Deep Mutational Scanning (DMS) datasets. Bars represent the average Spearman correlation across all datasets, while individual points denote performance on specific DMS tasks.

parison. This choice favors neither architecture *a priori*: linear-time models have better asymptotic complexity, but Transformers with FlashAttention-2 achieve competitive throughput at 8k context (Dao, 2023).

We evaluate the ability of different architectures to maintain biological fidelity over extended contexts. We prompt the models and allow them to generate continuously for up to 20,000 tokens. This rigorous setting tests the models’ capacity to extrapolate well beyond their training context window and maintain structural coherence over long sequences.

We utilize two sets of prompts for evaluation: 1) The first set comprised ten sequence prompts, each corresponding to a distinct instance of the first gene of the ribosomal *S10* operon, where gene content and order is near universally-conserved across bacterial species (Table 6) (Coenye & Vandamme, 2005), and 2) representative sequences of the top seven most abundant genes in BaseData that have functional annotations derived from the KEGG database (Kanehisa & Goto, 2000). We measure coding density and sequence complexity (proportion of bases unmasked by DUST (Morgulis et al., 2006)) as a function of generated sequence length.

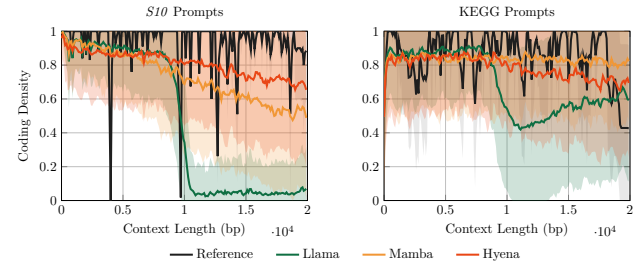


Figure 7. Coding density across context length for sequences generated with the *S10* and KEGG prompts, calculated for each non-overlapping 100 base pair (bp) interval. The shaded regions represent the standard deviation.

Figure 7 illustrates the coding density. We observe that while all models start with high fidelity, their performance diverges as the sequence length increases beyond the training context. Notably, the Llama model shows a sharp degradation in coding density immediately after exceeding its

training context of  $\sim 8k$  tokens. In contrast, the Mamba and Hyena models exhibit a much more gradual decline, suggesting better extrapolation capabilities for these architectures in the absence of explicit attention mechanisms over the full history.

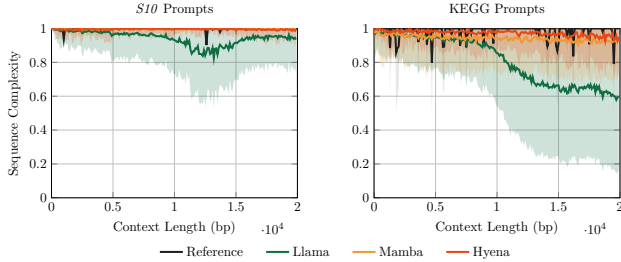


Figure 8. Sequence complexity (proportion of bases unmasked by DUST) across context length for sequences generated with the *S10* and KEGG prompts, calculated for each non-overlapping 100 bp interval.

Similarly, Figure 8 shows the sequence complexity. Maintaining complexity close to the reference is crucial for generating valid downstream biological hypotheses. Again, we see that the Transformer-based model struggles to maintain structural validity far beyond its training window, whereas the linear-time models degrade more gracefully.

## 5.2. Semantic coherence

While coding density and sequence complexity provide proxies for structural validity, they, alone, do not measure biological correctness. To address this, we evaluate whether the models generate the specific genes expected to follow the prompt beyond the context window of 8,192 tokens (i.e., the downstream genes of the *S10* and *spc* operons, which together span a highly conserved region of approximately 10,600 bp). Figure 9 shows the frequency of expected gene recovery across generated sequences.

The *S10* and *spc* operons are an ideal test case for semantic coherence because their gene content and order (*rplC*  $\rightarrow$  *rplD*  $\rightarrow$  *rplW*  $\rightarrow$  *rplB*  $\rightarrow$  ...) are highly conserved across bacteria (Coenye & Vandamme, 2005). While Coenye & Vandamme (2005) document variation in gene order across some phyla, the core *S10*-*spc* operon architecture remains one of the most stable genomic features in prokaryotes. A model that has learned the “grammar” of bacterial genomes should predict these downstream genes in the correct order when given the upstream context, analogous to how a language model predicts semantically coherent continuations.

We observe a trade-off between structural stability and semantic coherence. Although Llama exhibits a sharper decline in coding density beyond its context window (Figure 7), it recovers the expected downstream genes more often than the linear-time architectures, with Mamba coming a close second (Figure 9). Notably, this semantic coherence

is maintained beyond the 8,192 token context window used in training. Conversely, Hyena, while maintaining high sequence complexity and coding density, generates sequences that are structurally valid (high coding density, correct GC content, see Figure 14) but semantically diverge as the annotated genes are not consistent with the expected *S10* and *spc* operon genes. We term this phenomenon “semantic divergence” rather than “hallucination” to distinguish it from factual errors: these sequences may represent valid bacterial genes, but they do not reflect the conserved biological logic.

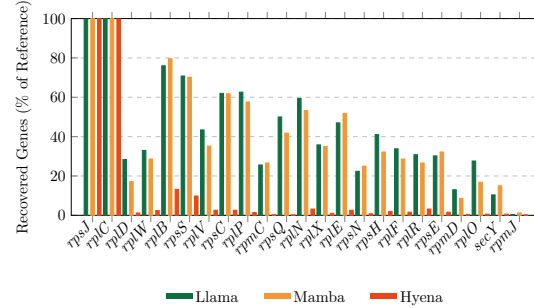


Figure 9. Percentage of generated contigs containing the expected genes for the *S10* benchmark in order of appearance.

## 6. Scaling to 28B parameters

Based on the comparative analysis presented in Section 4 and Section 5, we selected the Llama architecture for our large-scale training run. While linear-time architectures like Mamba and Hyena offer attractive inference properties and superior structural extrapolation beyond the training window, our results demonstrate that, beyond a certain scale, the Transformer architecture consistently outperforms them in metrics most critical for biological understanding; scaling efficiency on complex signals, zero-shot fitness prediction, and semantic coherence. Consequently, we proceeded to scale the Llama architecture to 28 billion parameters.

We emphasize that the 28B model serves primarily to validate our scaling law extrapolation rather than to establish state-of-the-art benchmarks. Comprehensive downstream evaluation of the final model is beyond the scope of this methodology-focused paper; our goal is to provide the systematic analysis and architectural insights that guided the model’s development, which we believe will be valuable to practitioners making similar design decisions.

### 6.1. EDEN-28B: validating the law

Guided by the scaling laws established in Section 4, we trained the flagship EDEN-28B model on the full 9.7 trillion token corpus. As illustrated in Figure 10, the final performance of EDEN-28B aligns closely with the predicted loss, validating the power-law extrapolation over orders of magnitude in scale. The trajectory from 100M to 28B parameters

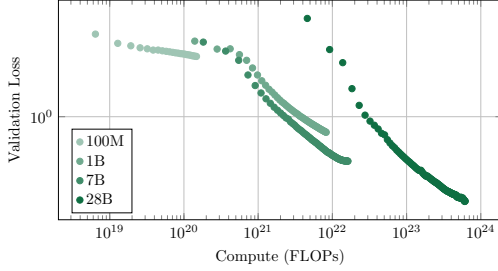


Figure 10. Scaling behavior of EDEN models ranging from 100M to 28B parameters.

exhibits a remarkable consistency, with the validation loss decreasing predictably as a function of cumulative FLOPs. This smooth scaling behavior confirms that the model architecture effectively utilizes the increased compute budget and has not yet reached a point of diminishing returns given the vast size of the BaseData corpus. This predictability provides a robust blueprint for future scaling, suggesting that further gains can be achieved with even larger models and broader data distributions.

## 6.2. The trade-off between structure and semantics

Contrary to the prevailing trend towards linear-time models for long sequences, our empirical results at the trillion-token scale favor the Transformer architecture (see Table 1). We interpret this through the lens of the *semantic coherence* vs. *structural stability* trade-off observed in Sections 4 and 5.

Table 1. Performance per task across each 7B architecture.

| Task                                          | Llama        | Mamba | Hyena       |
|-----------------------------------------------|--------------|-------|-------------|
| Mean pLDDT                                    | <b>78.99</b> | 72.65 | 70.82       |
| Mean TM-score                                 | <b>0.83</b>  | 0.73  | 0.73        |
| Domain architecture recovery (%)              | <b>94.00</b> | 73.33 | 69.33       |
| Mean perplexity across rare genes             | <b>2.47</b>  | 2.55  | 2.52        |
| Mean spearman correlation (RNAGym)            | <b>0.339</b> | 0.289 | 0.325       |
| Long context - mean masking rates             | 15           | 4.2   | <b>2.33</b> |
| Long context - mean coding density            | 0.6          | 0.78  | <b>0.81</b> |
| Long context - mean number of ORFs with Pfams | <b>0.4</b>   | 0.36  | 0.2         |

We interpret our results through the structural trade-off between context compression and precise retrieval. Linear-time architectures (Mamba, Hyena) achieve efficiency by compressing context into a fixed-size recurrent state. We hypothesize that while this compression is sufficient for maintaining global statistics, it creates a bottleneck when the task requires retrieving high-fidelity information from the distant past.

This limitation is particularly acute in biology, where sequence logic often relies on long-range dependencies, such as a distal regulatory element precisely matching a specific promoter - rather than aggregate properties. This interaction resembles a “retrieval” operation, which linear compression struggles to resolve.

Our empirical benchmarks strongly support this hypothesis. While both Mamba and Hyena excel at capturing global structural properties like coding density, they falter on tasks

requiring precise semantic resolution: Hyena struggles to reconstruct specific genetic identities, and Mamba significantly underperforms on zero-shot fitness prediction. In contrast, Llama’s full attention mechanism maintains lossless access to the entire history, enabling the precise token-to-token attention required to excel at these information-dense tasks. Therefore, for the purpose of a foundation model intended to capture high-fidelity biological representations, the Transformer’s superior semantic performance justifies its computational cost. Importantly, as shown in Table 5, the FLOPs cost of Llama is comparable to or lower than the linear-time alternatives at equivalent parameter counts: at 7B parameters, Striped Hyena requires  $\sim 0.9\times$  the FLOPs of Llama, while Mamba-Hybrid requires  $\sim 2.2\times$ . This means the Transformer does not incur a significant compute penalty relative to these hybrid architectures. Furthermore, with the advent of hardware-aware optimizations like FlashAttention-2 (Dao, 2023), the throughput gap between Transformers and linear models has narrowed significantly on modern hardware, making the Transformer’s superior modeling performance a viable trade-off for scaling.

## 7. Limitations and practical considerations

While EDEN demonstrates strong scaling and generation capabilities, it is not without limitations. The optimal architecture depends on the intended application: the quadratic complexity of the Transformer architecture imposes inference costs compared to linear-time models, potentially limiting deployment in resource-constrained environments; conversely, for tasks requiring ultra-long context (e.g., chromosome-scale analysis), linear-time models may be preferable despite their semantic limitations. For tasks requiring precise biological reasoning (e.g., fitness prediction, operon completion), our results suggest Transformers are the better choice at current scales. Additionally, while our results favor Transformers for semantic accuracy, they exhibit limited structural extrapolation beyond the training context compared to linear-time architectures. Validating generative models for “microbial dark matter” remains challenging; while we verify properties like synteny and coding density, true functional validation requires wet-lab characterization. Our scaling law analysis is based on four Llama scales (100M, 300M, 1B, 7B) and three scales per architecture for the cross-architecture comparison (100M, 1B, 7B), with bootstrapped perplexities over six evaluation seeds at each scale. Additional intermediate scales would further strengthen the statistical power of our exponent estimates; pre-training a denser ladder of models was not feasible within the compute budget and timeline of this study, and we leave the construction of a higher-resolution scaling grid to future work. We nonetheless believe the present analysis is of value to the community: it provides a reproducible, compute-aware blueprint for architecture selection in ge-

omic foundation models. Finally, our current model is trained solely on nucleotide sequences, potentially missing regulation encoded in epigenetic modifications or 3D chromatin structure. We also acknowledge that our primary dataset is proprietary, limiting full reproducibility; however, we provide a complete comparison on the public OpenGenome2 dataset, demonstrating that our key findings are reproducible on publicly available data. See Appendix B for more details on BaseData.

## 8. Conclusion

In this work, we introduced the design and training methodology behind the EDEN family of foundation models for metagenomics. By rigorously benchmarking architectures, we demonstrated that Transformers offer superior scaling efficiency and semantic coherence compared to linear-time alternatives, while the latter excel at structural extrapolation - a trade-off practitioners should consider based on their application requirements. We derived quality-aware scaling laws highlighting the critical role of data curation in model performance.

Crucially, we argue that evaluation of biological foundation models must evolve beyond perplexity. While statistical likelihood drives optimization, it is an imperfect proxy for biological plausibility and functional utility. The true impact of models like EDEN will be determined by their performance on downstream tasks requiring synthesis of complex evolutionary rules - predicting variant fitness, designing functional pathways, or engineering microbial communities. The field must prioritize functionally-grounded benchmarks that distinguish models capturing causal biological mechanisms from those merely memorizing sequence patterns.

## Impact Statement

This paper presents a large-scale foundation model for genomics, EDEN, which has the potential to accelerate therapeutic design and biological engineering. While the capabilities of such models offer significant benefits for human health - ranging from novel antibiotics to gene therapies - they also carry potential dual-use risks. We acknowledge the importance of biosafety in the development and deployment of generative biological models. As the field advances, we advocate for the continuous development of safety frameworks that evolve alongside model capabilities to ensure that these powerful tools are used for the benefit of humanity.

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## A. EDEN architecture details

All model architectures were implemented using the NVIDIA BioNeMo framework, which is built upon NeMo 2.0 and Megatron-LM. For the Llama architecture specifically, we utilized the Llama 3.1 implementation.

Table 2 provides the detailed configuration for the Llama models at different scales.

Table 2. Architecture configurations for Llama models at different scales.

| Configuration    | 100M   | 300M   | 1B     | 7B     | 28B    |
|------------------|--------|--------|--------|--------|--------|
| Layers           | 12     | 20     | 24     | 32     | 48     |
| Hidden dimension | 768    | 1024   | 2048   | 4096   | 6144   |
| Attention heads  | 12     | 16     | 16     | 32     | 48     |
| KV heads (GQA)   | 12     | 4      | 16     | 8      | 8      |
| FFN dimension    | 2048   | 4096   | 5632   | 11008  | 16384  |
| RoPE base freq   | 500000 | 500000 | 500000 | 500000 | 500000 |

Table 3 provides the detailed configuration for the Mamba models at different scales Table 4 provides the detailed configuration for the Hyena models at different scales For the hybrid architectures (Mamba2-Hybrid and Striped Hyena), we follow the configurations from [Brix et al. \(2025\)](#), where attention layers are interleaved with Mamba or Hyena layers at a ratio of approximately 1:7 (one attention layer per seven recurrent/convolutional layers).

## B. Data

The metagenomic data used in this work was assembled while aiming at increasing taxonomic and geographic diversity relative to commonly used public corpora for training biological language models. The primary dataset collection and training data curation has been driven by consent considerations to ensure it is fair to stakeholders (public data largely ignores that) including an incentive structure through benefit sharing. Each of the 9.7 trillion tokens in pretraining of EDEN family can be traced back to consent of stakeholders.

**Filtering criteria.** We retain contigs (contiguous sequences of DNA assembled from many shorter reads) only if they exceeded 2 kb in length and exhibited a predicted gene density greater than 20%. To ensure high-confidence sequences, we required a minimum mean sequencing depth of 4X. Low-complexity contigs shorter than 10 kb and containing more than 50% low-complexity sequence, as quantified with DUSTmasker (v2.15.0), were removed from training ([Morgulis et al., 2006](#)). Finally, for safety reasons, sequences with significant homology to known eukaryotic viruses were excluded to focus on prokaryotic and phage diversity.

Existing open datasets (e.g., OMG, OpenGenome, OpenGenome-2) often inherit similar limitations, including substantial redundancy and skewed representation across taxa and sampling locations. For example, sequences in major public repositories are heavily concentrated in a small number of species (e.g., in SRA, a large fraction of deposited sequence volume is attributable to a handful of organisms). The dataset used here is designed to mitigate these biases by sampling broadly across underrepresented lineages. As an example, it includes sequences from 1M species-level groups that are not represented in standard reference collections. By reducing redundancy and taxonomic imbalance, we expect our analysis to provide more robust scaling-law estimates than prior studies that rely on more biased sequence collections.

Table 3. Architecture configurations for Mamba models at different scales.

| Configuration    | 100M | 1B    | 7B    |
|------------------|------|-------|-------|
| Layers           | 20   | 24    | 52    |
| Hidden dimension | 768  | 2560  | 4096  |
| Attention heads  | 16   | 16    | 32    |
| KV heads (GQA)   | 8    | 8     | 8     |
| FFN dimension    | 4096 | 10240 | 21504 |
| SSM groups       | 8    | 8     | 8     |
| State dimensions | 64   | 128   | 128   |
| Head dimensions  | 96   | 160   | 64    |

Table 4. Architecture configurations for Hyena models at different scales.

| Configuration    | 100M  | 1B    | 7B    |
|------------------|-------|-------|-------|
| Layers           | 11    | 25    | 32    |
| Hidden dimension | 768   | 1920  | 4096  |
| Attention heads  | 24    | 15    | 32    |
| FFN dimension    | 3072  | 5120  | 11008 |
| Short filter     | 64    | 128   | 256   |
| Medium filter    | 64    | 128   | 256   |
| Long filter      | 768   | 1920  | 4096  |
| RoPE base freq   | 10000 | 10000 | 10000 |

### C. Training details

All models were trained using the AdamW optimizer with  $\beta_1 = 0.9$ ,  $\beta_2 = 0.95$ , and a weight decay of 0.01. We used a cosine learning rate schedule with 2500 warm-up steps, reaching a peak learning rate of  $3 \times 10^{-5}$  for all models and decaying to  $6 \times 10^{-7}$  over 640920 steps. We stopped training once the target token count was reached, and used the resulting checkpoint for evaluation. Training was conducted on NVIDIA H200 GPUs using bfloat16 mixed precision with FP8 hybrid mode. To support model scaling, we utilized Tensor Parallelism (TP=2) and Sequence Parallelism across all runs. As detailed in Table 5, the 100M and 1B models used a global batch size of 2048 distributed across 16 GPUs, while the 7B models utilized a batch size of 384 across 48 GPUs. The largest 28B model was scaled to 1008 GPUs across 126 nodes.

Table 5. Architecture scaling and training throughput summary

| Architecture | #Params | #FLOPs                 | Global batch | #Eff. tok/batch | Total tokens processed |
|--------------|---------|------------------------|--------------|-----------------|------------------------|
| Mamba        | 100M    | $0.56 \times 10^{21}$  | 2048         | 16777216        | 356515840000           |
|              | 1B      | $5.87 \times 10^{21}$  | 2048         | 16777216        | 356515840000           |
|              | 7B      | $36.75 \times 10^{21}$ | 384          | 3145728         | 349965385728           |
| Hyena        | 100M    | $0.28 \times 10^{21}$  | 2048         | 16777216        | 356515840000           |
|              | 1B      | $2.64 \times 10^{21}$  | 2048         | 16777216        | 356515840000           |
|              | 7B      | $14.33 \times 10^{21}$ | 384          | 3145728         | 349965385728           |
| Llama        | 100M    | $0.11 \times 10^{21}$  | 2048         | 16777216        | 356515840000           |
|              | 300M    | $0.63 \times 10^{21}$  | 2048         | 16777216        | 293584502784           |
|              | 1B      | $2.32 \times 10^{21}$  | 2048         | 16777216        | 356515840000           |
|              | 7B      | $16.36 \times 10^{21}$ | 384          | 3145728         | 349965385728           |

### D. Training and validation loss trajectories

We monitor the training stability and convergence through training and validation loss trajectories. Figure 11 shows the loss curves for the 100M/1B and 7B models, respectively. The 100M and 1B models (Figure 11a) were trained with a global batch size of 2048 sequences, while the 7B models (Figure 11b) used a batch size of 384 sequences. While all architectures exhibit consistent scaling behavior, we observe a shift in relative performance at scale: at the 7B parameter regime, the Llama model achieves the lowest validation loss, surpassing the recurrent architectures that performed comparably or better at smaller scales.

### E. Additional benchmarking results

In addition to the aggregate TM-score metrics presented in the main text, we provide a more granular view of structural plausibility in Figure 12. This breakdown by gene and model highlights that while all architectures are capable of generating plausible structures, Llama-based models exhibit greater consistency, with fewer low-quality generations across all three genes.

Figure 13 shows the relationship between structural accuracy (TM-score) and sequence identity to the reference. Notably, Llama models maintain high TM-scores even at lower sequence identities, suggesting they generalize structural constraints

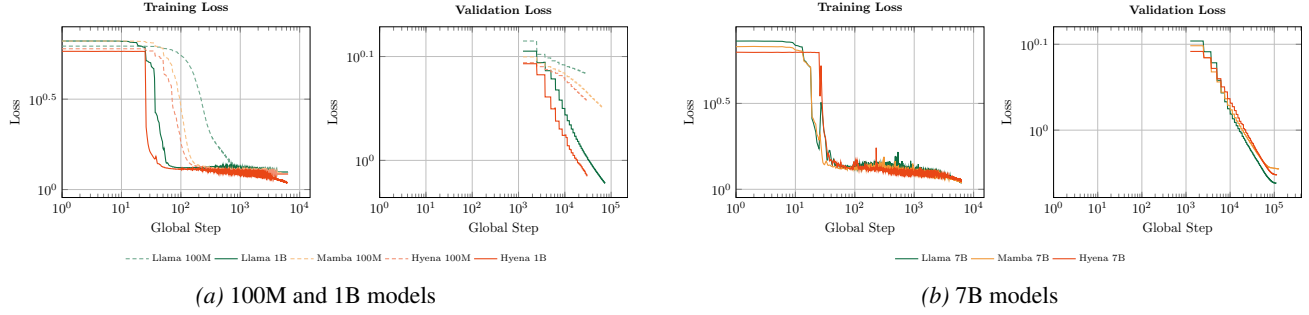


Figure 11. Training and Validation Loss Trajectories. (a) 100M and 1B models (dashed lines for 100M, solid for 1B). (b) 7B models.

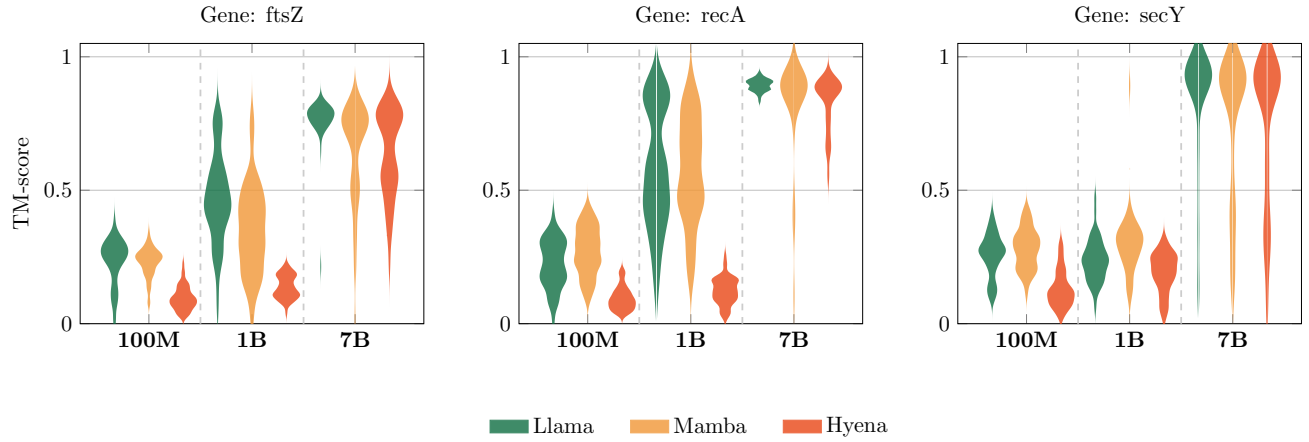


Figure 12. Distribution of TM-scores by gene and model. The violin plots highlight that while all models can generate plausible structures, Llama models exhibit greater stability with fewer low-quality generations.

better than Mamba and Hyena architectures.

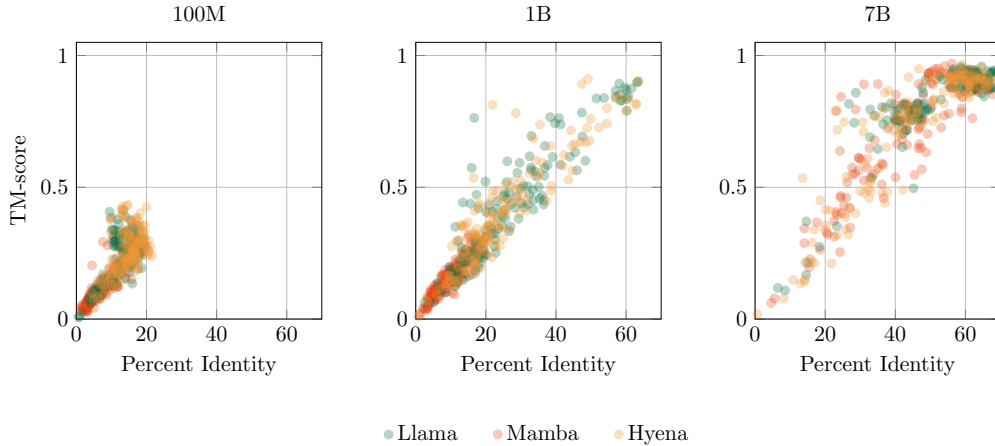


Figure 13. TM-score vs. Percent Identity. Larger models (7B) cluster in the upper-right quadrant. Llama models maintain high structural accuracy even at lower sequence identities, suggesting better generalization.

## F. Long-context generation metrics

In this section, we provide a detailed breakdown of the long-context generation performance across different models. We evaluate the models on their ability to generate biologically plausible sequences that extend beyond the typical training context window, using a variety of metrics to assess sequence quality, including ORF length, coding density, GC content, and functional annotation rates.

Table 6. The accession numbers and coordinates of the *S10* prompts used in the long-context generation evaluations.

| RefSeq accession | Genomic coordinates (start, end) | Orientation        |
|------------------|----------------------------------|--------------------|
| CP026387         | 2875037, 2876036                 | Forward            |
| CP028915         | 877336, 878335                   | Forward            |
| NC_000913        | 3452271, 3453270                 | Reverse complement |
| NC_003197        | 3595538, 3596537                 | Reverse complement |
| NC_003198        | 4232972, 4233971                 | Forward            |
| NC_004547        | 4502338, 4501337                 | Reverse complement |
| NC_009436        | 4054414, 4053413                 | Reverse complement |
| NC_009792        | 4350639, 4351638                 | Reverse complement |
| NC_013592        | 4267630, 4268629                 | Reverse complement |
| NC_017390        | 3682075, 3683074                 | Reverse complement |

Figure 14 presents a comprehensive comparison of these metrics for sequences generated by Reference, Llama, Mamba, and Hyena models. The results highlight the differing capabilities of each architecture in maintaining long-range coherence and biological fidelity. While all models capture basic sequence statistics, significant divergences appear in more complex metrics such as coding density and functional annotation rates (PFAM/KEGG), particularly for the Llama architecture which struggles to maintain these properties over extended generation lengths compared to the hybrid and recurrent architectures.

## G. Impact of scale on context extension for Llama model

The comparisons above utilize models trained on a controlled budget of 350 billion tokens to isolate architectural inductive biases. While the 7B Llama baseline exhibits degradation in structural metrics beyond its 8,192-token window, we find that scaling to 28B parameters offers negligible improvement in this regard. As illustrated in Figure 15, the EDEN-28B model, despite being trained on 9.7 trillion tokens, suffers from a similar degradation in both sequence complexity and coding density once generation exceeds the training context window. This observation suggests that the “context gap” is not merely a function of model scale or training duration but likely an intrinsic limitation of the architecture’s ability to extrapolate

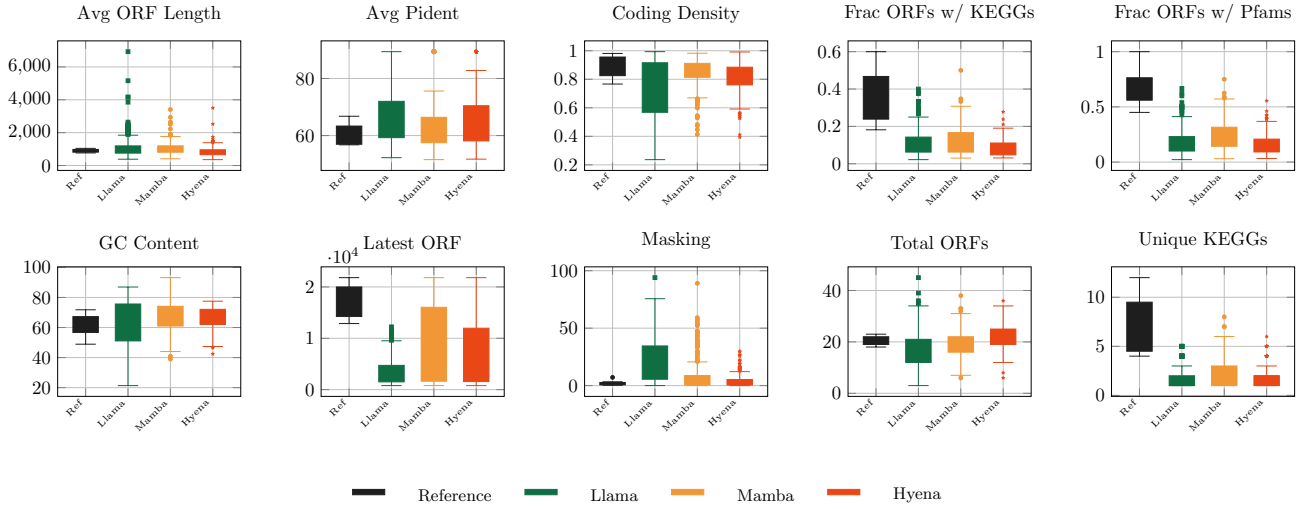


Figure 14. Detailed comparison of generation statistics for long-context KEGG prompts across Reference, Llama, Mamba, and Hyena models. The metrics include Average ORF Length, Average Percent Identity (Pident) to KEGG annotation, Coding Density, Fraction of ORFs with KEGG annotations/PFAM domains, GC Content, Latest ORF position, Masking rates, Total ORFs, and Unique KEGGs. Boxplots show the distribution of values for each metric, illustrating the variance and central tendency for each model.

beyond its positional training horizon.

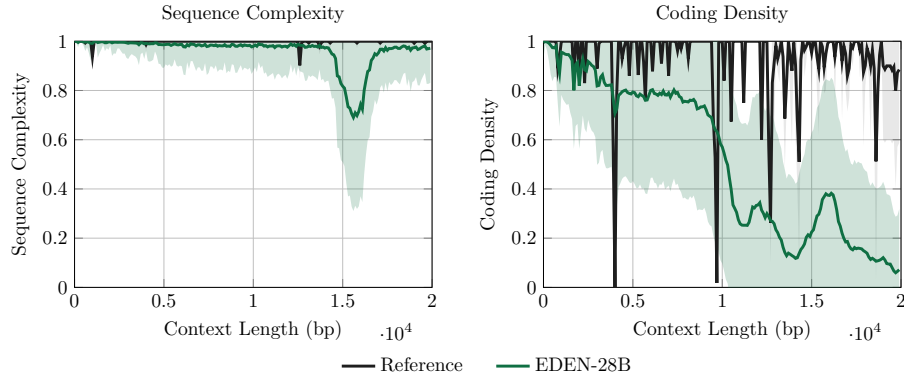


Figure 15. Sequence metrics for *S10* prompts. Comparison between the Reference (ground truth) and EDEN-28B model on (Left) Sequence Complexity and (Right) Coding Density. Despite the massive increase in scale, the EDEN-28B model fails to maintain metrics close to the reference distribution beyond the training context window ( $\approx 8,192$  bp), mirroring the limitations observed in smaller models.